AsmFEA - Getting started

Stress analysis of a whole SOLIDWORKS assembly, from a task pane, in a few clicks.

1 Find the pane | 2 Prepare | 3 Loads & fixtures | 4 Connections | 5 Run | 6 Results

1 Find AsmFEA in SOLIDWORKS

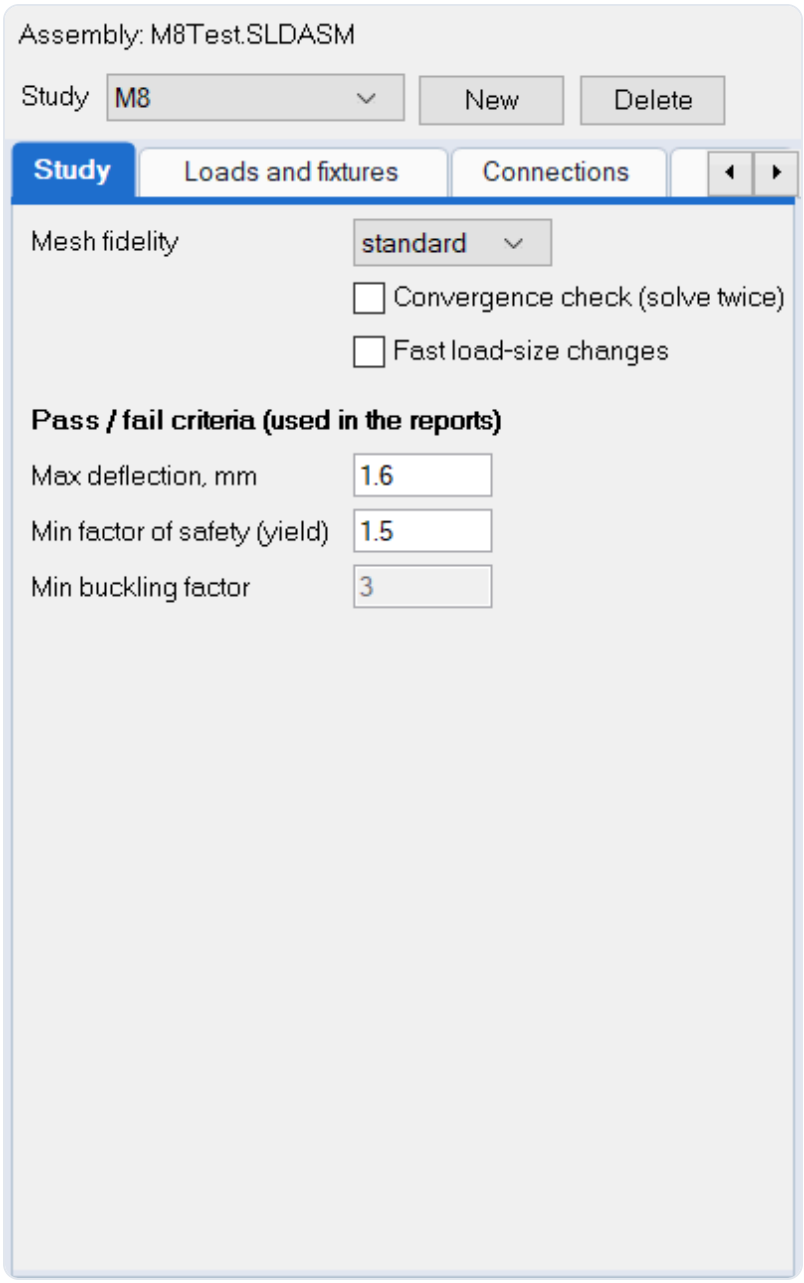
1. Start SOLIDWORKS (2018 or later; one of the versions you ticked in the installer). AsmFEA loads with it.
2. On the right of the window, the task pane has a new tab with the coloured hexagon icon (the one at the top of this page): click it to open AsmFEA.
3. If it is missing: **Tools > Add-ins**, tick **AsmFEA** in both columns, then OK.

2 Prepare the assembly and start a study

- Open the assembly and give every part a material (right-click the part > **Material**). A single part works too: its bodies are treated as the parts, and bodies that touch are joined.
- Click New in the pane and name the study. It is saved inside the assembly (or part).
- On the **Study** tab pick a mesh fidelity (*coarse* is quick for a first look) and check the pass/fail criteria the reports use. Changes are kept as you make them.
- If you expect to try several sizes of load, tick *Fast load-size changes*: each load is solved once, and changing only load sizes afterwards needs no new solve.

> Parts that touch are joined automatically. Mates tell AsmFEA where parts can slide or

> turn; weld beads are used too, and a sub-assembly is treated as one rigid piece.



The Study tab

3 Add fixtures and loads

1. Select one or more faces on the model.
2. On the **Loads and fixtures** tab click `Fixed` (held still) or `Pinned` (can rotate).
3. Loads belong to a **load case** (a set of loads solved together). Pick one at the top of the tab, or click `New load case`; every case is solved in the same run. Fixtures apply to all

cases. Tick **Include buckling in load case** to check a case for buckling too (runs take longer).

4. Select faces and click `Force` or `Pressure`, or click `Gravity` for self-weight. Give the force as X, Y and Z components; the resultant shows as you type.
5. For a force acting away from the faces (a lever arm, a torque), also give how far from the faces' centre it acts, in X, Y and Z (leave 0, 0, 0 for a force on the faces).
6. For fluid in a tank, select the wetted faces and click `Hydrostatic`. Give the fluid's density (water 1000 kg/m^3), the absolute pressure on its top (1 atmosphere, 0.101325 MPa, for an open tank; the air outside balances it), and where the top of the fluid is, in X, Y and Z from the faces' centre; a ball and triad marks it on the model. The pressure rises with depth below it; faces above it are not loaded.

Click a row in the list to highlight its faces on the model; a load also shows an arrow pointing at where it acts. Double-click a row to change it.

Assembly: M8Test.SLDASM

Study M8 New Delete

Study | Loads and fixtures | Connections

Load case LC1

Buckling 5 modes

New load case | Rename | Delete load case

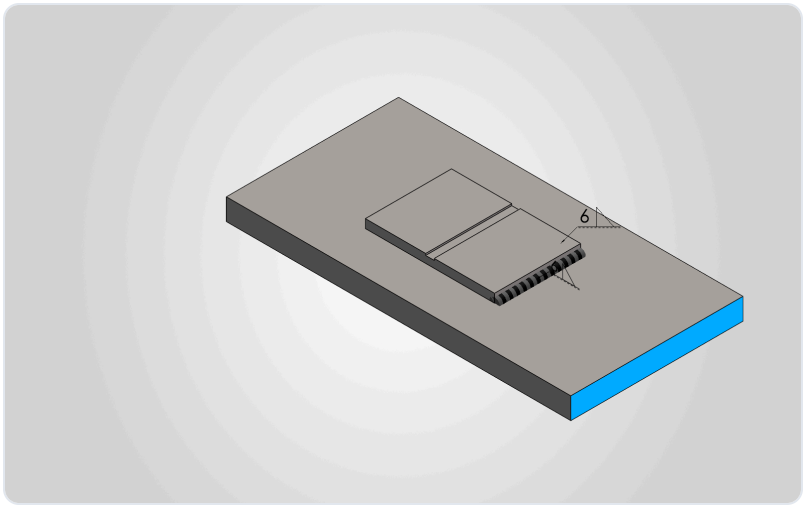
From the selected faces, add:

Fixed | Pinned | Force | Pressure

Gravity | Delete

Id	Kind	Type	Value	Faces
FX1	fixture	fixed		L3
LD2	load	gravity	0, -9810, 0 mm/s² (9810 mm/s²)	
LD1	load	force	0, -1000, 0 N (1000 N)	L1

Loads and fixtures

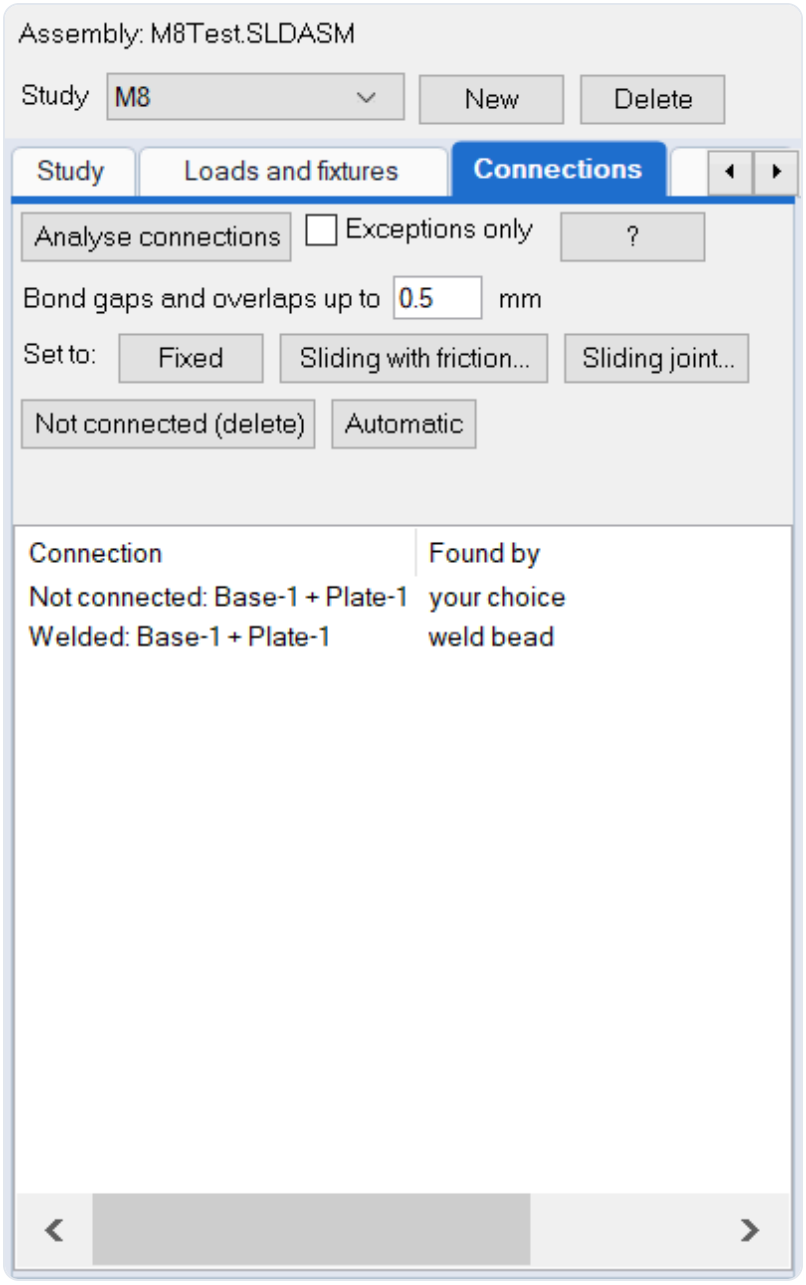


The fixed face highlighted in blue. This assembly is a plate on a base, held by a 6 mm weld bead along one edge.

4 Check how the parts connect

- Click `Analyse connections`: the list shows every place where parts meet and how they will be joined.
- To change one, select it and *Set to*: `Fixed` (bonded), `Sliding with friction...`, `Sliding joint...` or `Not connected`. `Automatic` undoes your choice.
- Weld beads appear as their own rows.

> Hover over any button for a tip. Double-click a connection or a warning, or press `?`, for a full explanation.

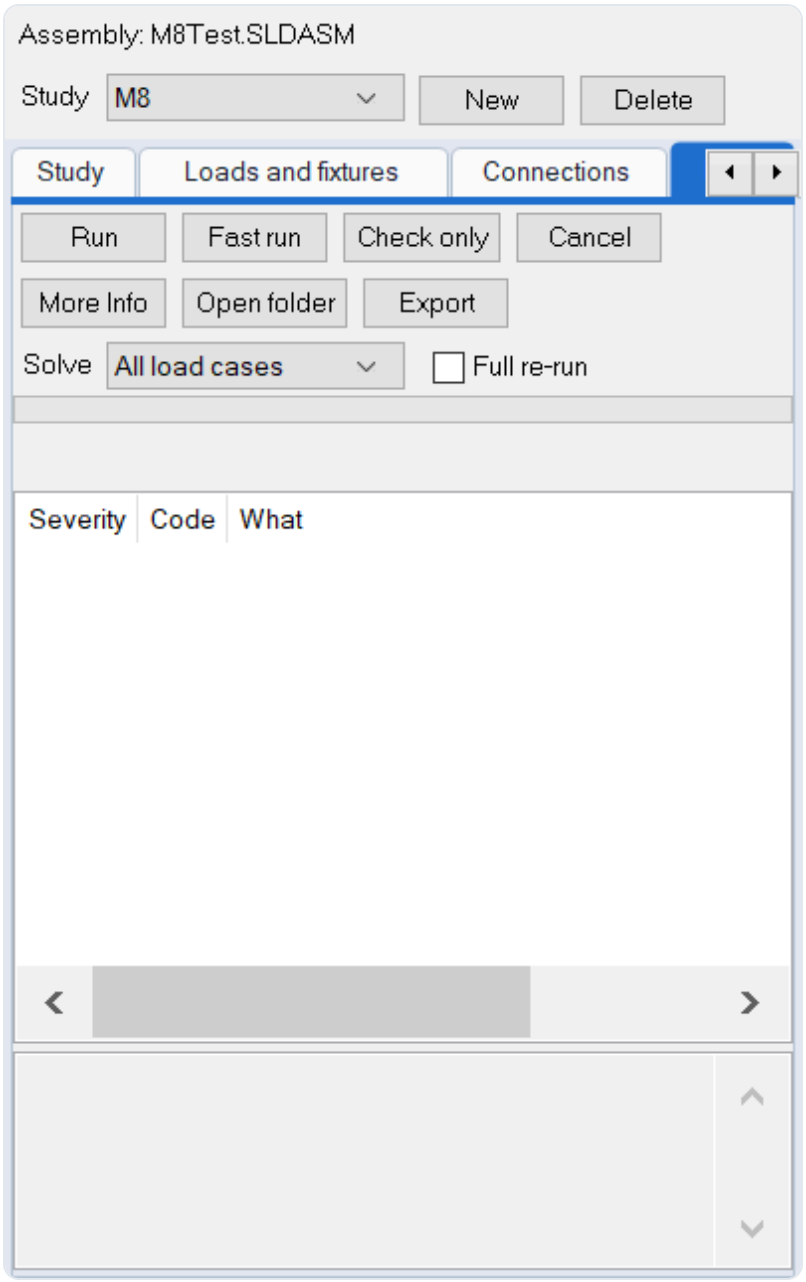


Connections, including a weld bead

5 Run

- On the **Run** tab click `Run`. The bar and the line below it show what is happening; a first run takes from under a minute to a few minutes.
- `Check only` just checks the set-up, quickly, without solving.
- `Fast run` is for trying design changes: a coarser model that shows where the problem is and whether a change helped, in a fraction of the time. Its results are marked **APPROXIMATE** and it writes no report: do a full `Run` at the end, before you rely on the numbers.

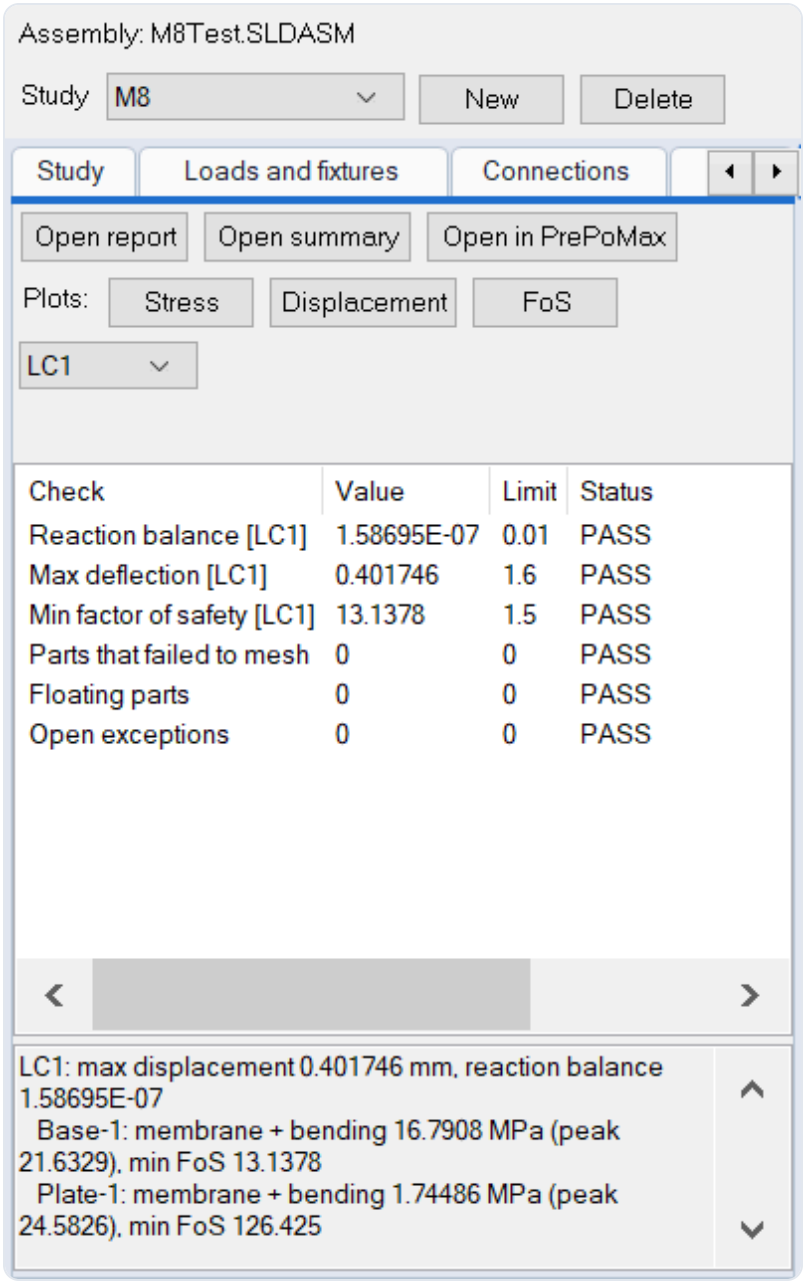
- A re-run redoes only what your changes need: with nothing changed it finishes in seconds, and a new pass/fail limit skips the solve. Tick *Full re-run* to redo everything.
- A model too large for CalculiX to fit in your computer's memory is no longer refused: AsmFEA solves it with its own hybrid solver, part by part, using the memory it has free. A note on the Run tab and in the report says when that happened.
- Anything that needs attention is listed: double-click it to see what it means and what to do.



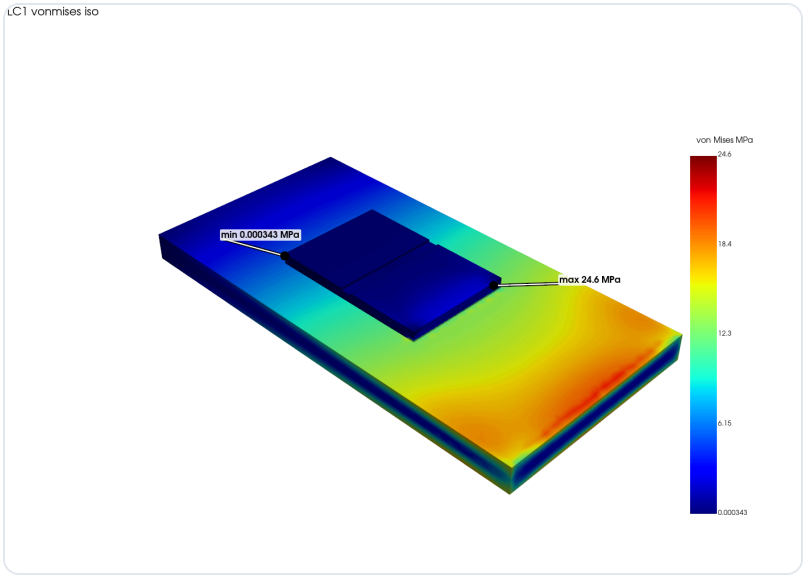
The Run tab

6 Look at the results

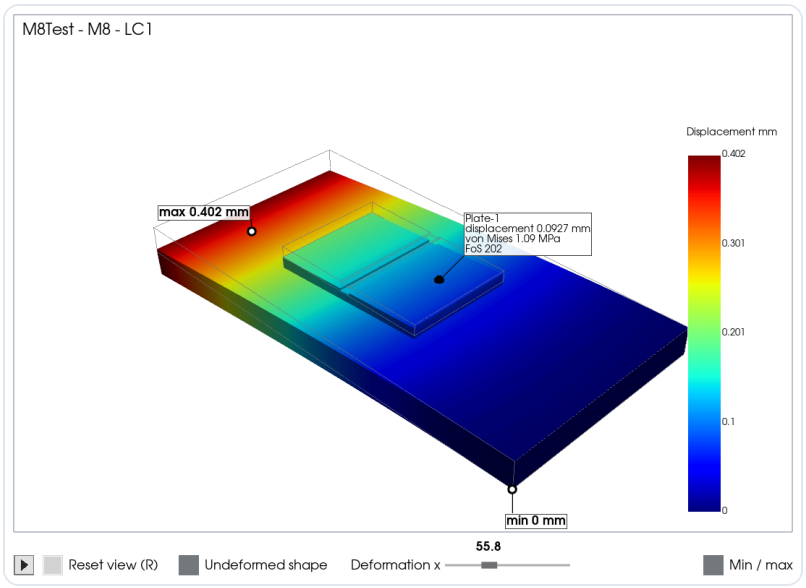
- The **Results** tab shows the pass/fail checks and the key numbers for each load case.
- `Stress`, `Displacement` and `FoS` open a 3-D window (their tooltips repeat what follows):
  - left drag to rotate, wheel to zoom, right drag to pan, as in SOLIDWORKS; **Reset view** (or R) goes back;
  - **click the model** to see the part and its stress, displacement and FoS at that point; click the background to clear it;
  - ▶ **play** ramps the load up and down so you can watch the stress build; click it again to stop at the full load;
  - **Deformation x** exaggerates the deformed shape, on every plot (0 shows it undeformed), and **Undeformed shape** adds a grey outline of the original;
  - **Min / max** shows the highest and lowest values; drag their labels anywhere.
- `Open report` gives the full Word report; `Open summary` a half-page summary. Both need a full run (they are greyed out after a fast run).
- The pictures follow the way you look at the model in SOLIDWORKS: turn it, then open the results again.


Assembly: M8Test.SLDASM
Study
M8
New
Delete
Study
Loads and fixtures
Connections
Open report
Open summary
Open in PrePoMax
Plots:
Stress
Displacement
FoS
LC1
Check
Value
Limit
Status
Reaction balance [LC1]
1.58695E-07
0.01
PASS
Max deflection [LC1]
0.401746
1.6
PASS
Min factor of safety [LC1]
13.1378
1.5
PASS
Parts that failed to mesh
0
0
PASS
Floating parts
0
0
PASS
Open exceptions
0
0
PASS
LC1: max displacement 0.401746 mm, reaction balance 1.58695E-07
Base-1: membrane + bending 16.7908 MPa (peak 21.6329), min FoS 13.1378
Plate-1: membrane + bending 1.74486 MPa (peak 24.5826), min FoS 126.425

Results



Stress (von Mises), with the highest and lowest values marked



The 3-D window: displacement with the undeformed outline, a clicked point and the min / max labels

7 Optional: PrePoMax, for a closer look

PrePoMax is a free program for exploring results in detail: section cuts, values at any point, animations. AsmFEA does not need it, but `Open in PrePoMax` on the Results tab opens the last run in it.

1. Download it from prepomax.fs.um.si/downloads (version 2.6.0 is the one AsmFEA was tested with).

2. It is a zip file: unzip it anywhere, for example `C:\PrePoMax`. There is nothing to install.
3. In AsmFEA click `Open in PrePoMax` and choose **Find PrePoMax.exe...** the first time; AsmFEA remembers it.

✓ How accurate is AsmFEA?

The verification report (also in the Start menu, and as a PDF in the AsmFEA folder) compares AsmFEA with classical hand calculations on four test cases (a cantilever, a beam under its own weight, two members joined end to end, and a buckling column) and checks the CalculiX solver against its own test examples and two NAFEMS benchmarks. Every comparison came within its acceptance criterion, the largest difference being 0.6 %.

Satisfy yourself of its accuracy for your own work by running your own tests against hand calculations or other trusted results before relying on it.

8 Keep your work

- The study is saved **inside the assembly**: save the assembly to keep it.
- Results and reports are in a folder next to the assembly called `<assembly name>_FEA`.
- Units are millimetres, newtons and MPa throughout.

> Always check its results with engineering judgement and a hand calculation before relying on them for anything that matters.

AsmFEA uses CalculiX (solver) and Gmsh (mesher), free software downloaded by the installer; see the licenses folder.